



THOMAS BADET

PhD

CONTACT

	MALE, 36 FRENCH
	RUE DU TERTRE, 4 2000 NEUCHÂTEL
	thomas.badet@hotmail.fr
	+41 76 683 51 06
	ORCID - 0000-0001-6130-441X SCHOLAR - i1iP7FYAAAAJ

TRAINING

21/12/2017	• PhD in Plant Microbe Interactions Paul Sabatier University - Toulouse III
06/2014	• Master degree in Plant Bioscience Paul Sabatier University - Toulouse III
09/2012	• Plant Microbe Interactions
06/2011	• Bachelor degree in Biology Rennes I University
09/2007	• Cellular Biology and Microbiology

SKILLS

MOLECULAR BIOLOGY

Plant and fungi DNA/RNA extraction
PCR, qRT-PCR, cloning

CELLULAR BIOLOGY

Agrobacterium-mediated transitory transgene
expression
Confocal microscopy and axiozoom

COMPUTATIONAL BIOLOGY

Genome/transcriptome mapping, variant calling
Genome-wide association
Genome-scale analysis

PERSONAL

Autonomy
Scientific planification

EXPERIENCES

present	• Maître-Assistant lecturer UNINE, Prof Vermeer's lab
01/04/2021	• Post-doctoral researcher UNINE, Prof Croll's lab
01/04/2025	• Genome evolution in fungi
01/04/2018	• PhD student LIPM INRA-CNRS, Dr Raffaele's supervision
28/02/2018	• Plant pathology
01/11/2014	• MASTER'S THESIS LIPM INRA-CNRS, Dr Raffaele's supervision
30/06/2014	• Quantitative Disease Resistance
01/11/2013	

COMMUNICATIONS

01/2026	• Phytopathology - Mycology meeting Invited speaker
11/2025	• Symposium on Fungal Genetics in IDF Invited speaker
03/2025	• ECFG17 meeting Attendee poster
01/2025	• Swiss Plant meeting Attendee talk
04/2024	• ICTE24 meeting Attendee poster
03/2023	• ECFG16 meeting Attendee poster
01/2023	• Swiss Plant meeting Attendee talk
06/2020	• Lausanne Genomics Days Invited talk
10/2020	• RIP2020 meeting Attendee / organizing committee
09/2019	• HOST-MICROBE GENOMICS meeting Attendee poster
07/2019	• SMBE meeting Attendee poster
09/2018	• EMBO Workshop-Experimental Approaches to Evolution and Ecology Attendee poster
11/2017	• HOST-MICROBE GENOMICS meeting Attendee talk
05/2017	• CELLULAR & MOLECULAR FUNGAL BIOLOGY GORDON RESEARCH Attendee poster
05/2016	• RESISTANCE network Attendee talk
06/2016	• PLANT HEALTH and ENVIRONMENT (SPE) PhD STUDENT MEETING Organizer
07/2014	• PLANT PHYSIOLOGY TEACHING

PUBLICATIONS

Root tip excision-induced exodermis lignification impacts lateral root emergence in *Brachypodium distachyon*.

Bellande K, De Jesus Vieira Teixeira C, D'Orlando A, Perez L, Sibout R, Roulin AC, Vermeer JEM and **Badet T**
New Phytologist 2026

High-throughput discovery of emerging antifungal resistance in crop pathogens.

Puccetti G, Flückiger D, Edel D, Delude C, Tralamazza SM, **Badet T**, Scalliet G and Croll D
bioRxiv 2025 (under revisions at Nature Microbiology)

A large European diversity panel reveals complex azole fungicide resistance gains of a major wheat pathogen.

Puccetti G, **Badet T**, Flückiger D, Edel D, Feurtey A, Delude C, Gluck-Thaler E, Torriani SFF, Scalliet G and Croll D
mBio 2025

A retrotransposon insertion upstream of Arabidopsis thaliana CRK8 receptor-like kinase modulates a trade-off between pathogen defense and salt tolerance.

Macquet J, **Badet T**, Carvalho-Silva P, Zaffuto M and Raffaele S
The Plant Journal 2025

A systematic screen for genetic factors underpinning transposon defense systems across the fungal kingdom

Badet T and Croll D
PLOS Biology 2025

An inversion polymorphism under balancing selection, involving giant mobile elements, in an invasive fungal pathogen

Hartmann FE, Rodriguez de la Vega RC, Demené A, **Badet T**, Vernadet J-P, Rougemont Q, Labat A, Snirc A, Stauber L, Croll D, Prospero S, Dutect C and Giraud T
Molecular Biology and Evolution 2025

Recent reactivation of a pathogenicity-associated transposable element is associated with major chromosomal rearrangements in a fungal wheat pathogen

Badet T, Moser Tralamazza S, Feurtey A and Croll D
Nucleic Acids Research 2024

A systematic screen for co-option of transposable elements across the fungal kingdom

Oggenfuss U, **Badet T** and Croll D
Mobile DNA 2024

Machine-learning predicts genomic determinants of meiosis-driven structural variation in a eukaryotic pathogen.

Badet T, Fouché S, Hartmann FE, Zala M and Croll D
Nature Communications 2021

Emergence and diversification of a highly invasive chestnut pathogen lineage across south-eastern Europe.

Stauber L, **Badet T**, Feurtey A, Prospero S and Croll D
eLife 2021

A population-level invasion by transposable elements triggers genome expansion in a fungal pathogen.

Oggenfuss U, **Badet T**, Wicker T, Hartmann FE, Singh NK, Abraham LN, Karisto P, Vonlanthen T, Mundt CC, McDonald BA and Croll D
eLife 2021

Rapid sequence evolution driven by transposable elements at a virulence locus in a fungal wheat pathogen.

Kumar Singh N, **Badet T**, Abraham L and Croll D
BMC Genomics 2021

The rise and fall of genes: origins and functions of plant pathogen pangenomes

Badet T and Croll D
Current Opinion in Plant Biology 2020

A 19-isolate reference-quality global pangenome for the fungal wheat pathogen *Zymoseptoria tritici*.

Badet T, Oggenfuss U, Abraham L, McDonald BA and Croll D
BMC Biology 2020

Stress-Driven Transposable Element De-repression Dynamics and Virulence Evolution in a Fungal Pathogen.

Fouché S, **Badet T**, Oggenfuss U, Plissonneau C, Francisco CS and Croll D
Molecular Biology and Evolution 2019

Expression polymorphism at the *ARPC4* locus links the actin cytoskeleton with quantitative disease resistance to *Sclerotinia sclerotiorum* in *Arabidopsis thaliana*.

Badet T, Léger O, Barascud M, Voisin D, Sadon P, Vincent R, Le Ru A, Balagué C, Roby D and Raffaele S
New Phytol. 2019

Parallel evolution of the *POQR* prolyl oligo peptidase gene conferring plant quantitative disease resistance.

Badet T, Voisin D, Mbengue M, Barascud M, Sucher J, Sadon P, Balagué C, Roby D and Raffaele S
PLoS Genetics 2017

Codon optimization underpins generalist parasitism in fungi.

Badet T, Remi P, Mbengue M, Navaud O, Derbyshire M, Oliver PR, Barbacci A and Raffaele S
eLife 2017

Emerging Trends in Molecular Interactions between Plants and the Broad Host Range Fungal Pathogens *Botrytis cinerea* and *Sclerotinia sclerotiorum*.

Mbengue M, Navaud O, Peyraud R, Barascud M, **Badet T**, Vincent R, Barbacci A and Raffaele S
Frontiers in Plant Science 2016

Common protein sequence signatures associate with *Sclerotinia borealis* lifestyle and secretion in fungal pathogens of the *Sclerotiniaceae*.

Badet T, Remi P and Raffaele S
Frontiers in Plant Science 2015

Resistance to phytopathogens *e tutti quanti*: placing plant quantitative disease resistance on the map.

Roux F, Voisin D, **Badet T**, Balagué C, Barlet X, Huard-Chauveau C, Roby D and Raffaele S
Molecular Plant Pathology 2014

SIDE EXPERIENCES



FUNGI PICKING



ASSOCIATIVE WORK



MUSIC



RADIO (SCIENTIFIC VULGARISATION)

TEACHING

MSc

Plant Pathology
Bioinformatic tools

BSc

Introduction to Phytopathology

REFEREES

SYLVAIN RAFFAELE

Research Director INRA | LIPM
Castanet-Tolosan
sylvain.raffaele@inrae.fr
+335 61 28 53 26

DOMINIQUE ROBY

Research Director CNRS | LIPM
Castanet-Tolosan
dominique.roby@inrae.fr
+335 61 28 55 11

SABINE FILLINGER

Research Director INRA | BIOGER
Thiverval-Grignon
sabine.fillinger@inrae.fr
+331 30 81 45 65

DANIEL CROLL

Professor | UNINE, Neuchâtel
daniel.croll@unine.ch
+41 32 718 23 30

JOOP VERMEER

Professor | UNINE, Neuchâtel
josephus.vermeer@unine.ch
+41 32 718 22 17